

This Month's DVD Contents

This month's DVD is a collection of design and simulation software such as power simulator (PSIM) for power electronics and a multipurpose integrated development environment (IDE) for ARM microcontrollers for seamless control on application development

SNEHA AMBASTHA

RIDE

Raisonance IDE, or RIDE, not only provides a platform for writing but controls compilation and debugging of the application code. This particular toolset is for ARM microcontrollers and is capable of rebuilding the peripheral configuration to automatically generate the resulting C source code. It is also an integrated simulator, a syntax highlighting code editor that can debug high-level languages with a rich range of views (such as locals, symbols and registers).

PSIM

PSIM is one of the fastest simulators for power electronics. It retains simulation accuracy during fast simulation because of which it is considered efficient for simulating converter systems of any size and for performing multiple-cycle simulations. Due to fast simulation, design cycles are significantly shortened and repetitive simulation runs are possible. PSIM can control circuit simulation in various forms such as analogue circuits, z-domain transfer function block diagrams, custom C codes and MATLAB/Simulink, among others.

LASI

LAYout System for Individuals or Individualists, or LASI, is a PC based computer aided design software primarily created for Windows operating system (OS). It is a layout and design tool for full-custom very large scale integration microelectronic circuit designs that include MEMS, integrated circuits (ICs), discrete devices, PC boards, schematics, different project documentation designs and other technologies. It can create and arrange polygons corresponding to regions like metal interconnect and

drain to create each device in the IC or any other design.

Logisim

Logisim is an open source educational tool for designing and simulating digital logic circuits. It makes learning basic concepts of logic circuits easy with its simple toolbar interface and simulation

of circuits. It has the capacity to build larger circuits from smaller sub-circuits and the capability to draw bundles of wires with just a single drag of a mouse. Logisim is used for entire simulation of central processing units for educational purposes.

Micro-Cap 11

Micro-Cap 11 is an integrated schematic editor and a mixed analogue or digital simulator with an interactive sketching and simulation environment for electronics engineers. It is capable of blending a modern and intuitive interface with robust numerical algorithms to enable ease of use and unparallelled levels of simulation power. It has an integrated circuit optimiser with multiple optimisation methods.

VeeCAD

VeeCAD is a freeware layout editor that takes a pre-generated schematic netlist and assists in building a layout on strip (Veroboard), matrix, donut, tripad, Atarado Smart and other prototyping boards. It can import netlists, define component footprints, position components and check wiring and connectivity. It is used for prototyping and building small quantities, hence, saves hours in planning layouts on graph paper. VeeCAD makes proto-

Popular resources

- 7Zip. Version v.9.38 (file archiver)
- Java Runtime Environment. Version 8.0.400.26 (runs programs written in Java programming language)
- VLC. Version 2.2.0 (media player)
- Opera. Version 28.0.1750.48 (Web browser)
- Free Download Manager. Version 3.9.4 (file download manager)
- Thunderbird. Version 31.1.0 (email application)
- Mozilla Firefox. Version 36.0.4 (Web browser)
- Apache Open Office. Version 4.1.1 (open office)
- Zillya AVEngine SDK (make-your-own antivirus)

typing practical, where one can draw the schematic in the usual schematic editor and use the same schematic with printed circuit board (PCB) editor.

OpenViBE

Open Virtual Brain Environment (OpenViBE) is a software platform for designing, testing and using brain-computer interfaces (BCIs). It enables communication between a human brain and a computer. BCIs are communication systems that allow users to send commands to computers through brain activities. It is open source and is available freely for real-time processing of brain signals.

FreeEDA

Previously known as OScad, FreeEDA is an open source electronics design and automation tool used for circuit design, PCB design, simulation and analysis. It is an integrated tool built using other open source software such as KiCAD, Ngspice and Schlab. It offers similar capabilities and ease of use as any equivalent proprietary software used for PCB design, schematic creation and simulation. It runs on many OSes including Ubuntu Linux, Windows XP and Windows 7. ●

The author is a technical correspondent at EFY